

Building Symmetrical Reality Systems for Cooperative Manipulation

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ABSTRACT

Humans and virtual agents (usually embodied in the physical world as physical robots) could coexist within the shared environment, which naturally creates a mixed reality environment of the physical and virtual worlds. In this paper, We analyze the characteristics of symmetrical reality with two different perception centers, and all of the mixed forms of physical and virtual reality can be treated as special cases of symmetrical reality. The experiment of folding clothes is implemented to demonstrate the significance of symmetrical reality for human-robot interaction, which also shows the general pipeline of building the symmetrical reality system with two minds (*i.e.*, the human mind and agent mind) involved.

Index Terms: Human-centered computing—Human-computer interaction (HCI)—HCI theory, concepts and models; Human-centered computing—Human-computer interaction (HCI)—Interaction paradigms—Collaborative interaction

1 INTRODUCTION

Human-robot interaction can go beyond purely physical space, and virtual environments have been widely adopted for embodied intelligence [4, 5]. Traditionally, robots are treated as purely physical and mechanical entities with software. However, as artificial intelligence is approaching a higher level, robots might think and act like humans in the future. Especially, artificial general intelligence (AGI) [1] gives robots the ability to have complicated communication with humans. This causes a paradigm shift in human-robot interaction [9]. Specially speaking, artificial intelligence will create a virtual agent that is located in an artificial virtual world, and the virtual agent could perceive and act in both the virtual and physical worlds. Meanwhile, the physical human could also perceive and act in both the physical and virtual worlds, which is similar to what the virtual agent does. As a consequence, the symmetrical structure of cognition [7] and interaction [6] is built, promoting the paradigm of human-robot interaction from one independent center (*i.e.*, the humans) to two independent centers (*i.e.*, the humans and the virtual agents).

To interpret the human-robot interaction within the aforementioned framework, two issues should be clarified. One is how an agent mind can be simulated, and the other is how the new paradigm of human-robot interaction work. For the first issue, to simulate a human-like virtual agent, it is necessary to build a human-like artificial mind. In recent years, artificial narrow intelligence only focuses on dealing with a specific problem, such as using neural networks for image recognition or playing games, and the ability to deal with one problem has outperformed the best performance of humans. But this is still far from what we expected. What the virtual agent should be equipped with should be artificial general intelligence, although it is still not realized. For the second issue, to know the features of the new paradigm of human-robot interaction, we should review some background of virtual-physical fusion. Symmetrical Reality (SR) [8] is proposed to unify all of the existing forms about the combination of the physical and virtual worlds, providing a better understanding of human-robot interaction. From the perspective of symmetrical

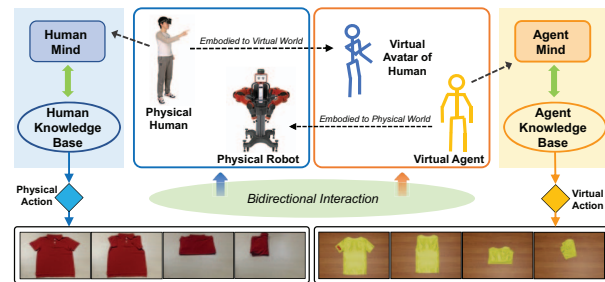


Figure 1: The explanation of symmetrical reality in the domain of human-robot interaction. The framework is focused on the symmetrical structure and the equivalent status of humans and virtual agents, which treats the physical robot as the physical embodiment of the virtual agent. The bidirectional interaction can promote human-robot collaboration.

reality, a robot is the physical embodiment of the virtual agent that is located in the virtual world, while humans can enter the virtual world with virtual embodiments to interact with other agents and environments.

In this paper, we leverage the concept of symmetrical reality and introduce how it could be applied to human-robot interaction tasks. Fig. 1 is the general description of a human-robot interaction task based on symmetrical reality. Our contributions are summarized as follows: i) The concept of symmetrical reality is leveraged to illustrate the symmetrical structure of humans and robots across the physical and virtual worlds; ii) We construct a task (folding clothes) simulator to realize human-robot cooperative tasks, and demonstrate how symmetrical reality can promote human-robot cooperation. Since the symmetrical reality framework is adopted to analyze the human-robot cooperative tasks, which treats the physical robot as the embodiment of the virtual agent, we directly use the “virtual agent” to replace the “robot” for convenience in the following sections.

2 EXPERIMENTS

To build an experimental platform for human-agent bidirectional interaction, we have adopted an optical see-through head-mounted display (OST-HMD) to present virtual information to humans. At the same time, the physical information is mirrored in the virtual world, so the virtual agent can monitor the real-time states of the physical objects. The above settings are typical configurations in symmetrical reality, *i.e.*, both humans and virtual agents can observe physical and virtual information.

For the experiment, we take the folding task as an example to demonstrate how symmetrical reality works, as shown in Fig. 2. This figure has shown the detailed bidirectional interaction process for physical humans and virtual agents. To bind with the features of SR, we treat the physics-based simulation shown in Fig. 2 as a simulated passive agent mind, which observes the human actions and makes determinations about the folding commands. In other words, a virtual simulator with stochastic properties could be regarded as a simulated mind to some extent. Based on this foundation, we conduct two experiments to show how the cooperation between human minds and agent minds helps improve the cooperative task.

In the given task, humans learn from self-demonstration and agent instruction, and the feedback generated from the former trial

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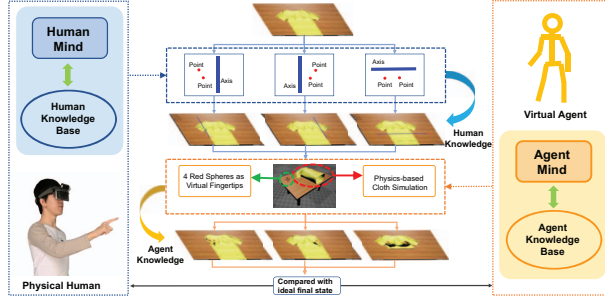


Figure 2: The cooperation process between humans and virtual agents in the task of folding T-shirts. The human first uses human knowledge to determine an action based on the observed state and then transfers the information about actions to the virtual world. The virtual agent receives the information of actions and implements the actions using the agent knowledge (*i.e.*, physics in the virtual world), and then transfers the result back to the physical human. Finally, the human can determine a new action according to what has been observed. During this interaction process, the physical human and the virtual agent both have their own minds and knowledge about the given task.

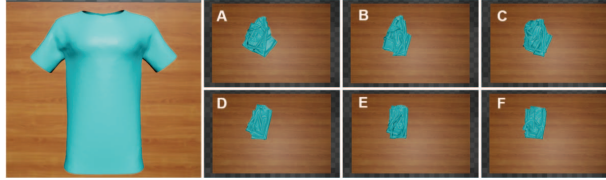


Figure 3: Results of folding the same clothes with only one method.

will instruct the latter trial. The feedback is generated according to how far the current folding state is from the ideal final state. The principle is similar to the backpropagation of errors in deep learning [3], and is also related to the policy gradient method in reinforcement learning [2].

2.1 Experiment I

2.1.1 The Same Clothes with the Same Folding Method

The first experiment is folding clothes in a virtual environment. The folding way is fixed to be {"right sleeve"→"left sleeve"→"fold torso vertically"→"fold torso horizontally"}. We record the folding results for every trial and ask 10 people to score the results. There are 6 trials (from A to F) for the same method, but the latter trial will be improved according to the result and feedback of the former trial. According to the ranking results regarding how well the T-shirt is folded, these 6 folding results will be assigned scores from 1 to 6. A higher score means a better result.

2.1.2 Result

Experiment I is designed to verify whether the symmetrical reality-based human-agent interaction could be beneficial for improving task performance. Fig. 3 has shown that different trials from A to F correspond to different folding results. Fig. 5 (left) indicates that the performance is being improved according to human scores.

2.2 Experiment II

2.2.1 The Same Clothes with Different Folding Methods

The second experiment is folding the same clothes in different ways. In this experiment, we demonstrate 3 different folding methods (from A to C) for one T-shirt. We record the folding result for every trial and ask 10 people to score the results. When the 3 methods are all finished, we ask people to rank the 3 folding results regarding how well the T-shirt has been folded.



Figure 4: Results of folding the same clothes with different methods.

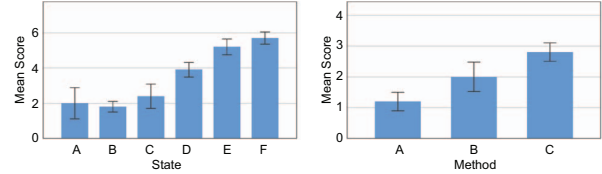


Figure 5: User evaluation results. (left) User evaluation of different folding results with one method. (right) User evaluation of different folding results with different methods.

2.2.2 Result

Experiment II is designed to verify whether the proposed framework could help to find the optimal solution for the given task. Fig. 4 has shown three different folding methods for the same T-shirt. Different folding methods will lead to different results. The result of the human score in Fig. 5 (right) has shown that the C method has the highest score, which means the C method is the best choice.

3 CONCLUSION

We have analyzed the reason why the symmetrical reality framework is promising in modeling the human-agent interaction across the physical and virtual worlds. Artificial intelligence is assumed to construct the mind of the virtual agent in the virtual world. Human-agent interaction has been expanded to the bidirectional interaction structure. Though the minds of agents are still far from independent compared to human minds, the agents can do lots of specific tasks in collaboration with humans. We have used experiments about folding clothes to prove the significance of SR in the domain of human-agent interaction, and the theory of symmetrical reality may benefit the study of human-agent interaction in the future.

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